Employee Database Management System using MySQL

1. Project Objective

The **Employee Database Management System** is a relational database project developed using **MySQL** to efficiently store, organize, and manage employee-related information. This project focuses on designing a well-structured database by implementing **Data Definition Language (DDL) commands** and **SQL constraints** to ensure data accuracy, consistency, and integrity.

The database includes tables such as Employees, Departments, and Locations, with relationships established using primary keys and foreign keys. Various constraints—including **Primary Key, Foreign Key, NOT NULL, UNIQUE, CHECK, and DEFAULT**—are applied to enforce business rules and maintain reliable data.

This project demonstrates fundamental database design concepts and serves as a practical example of building a robust relational database for employee management.

2. Problem Statement

- Design a structured relational database for managing employee information.
- Create database tables using MySQL **DDL commands** such as CREATE, ALTER, DROP, and TRUNCATE.
- Implement **constraints** to ensure data integrity and consistency.
- Establish relationships between tables using **Primary Keys** and **Foreign Keys**.
- Enforce validation rules using **NOT NULL**, **UNIQUE**, **CHECK**, and **DEFAULT** constraints.
- Build a scalable database structure that minimizes data redundancy through normalization.
- Develop a strong foundation in SQL database design and schema management for real-world business applications.

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3. Attribute (Column /Features) Details:

Table 1: Departments

Attribute Name	Data Type	Description
Department_id	Int (Primary Key)	Unique identifier assigned to each department
Department_ name	Varchar (100)	Name of the Department

Table 2: Locations

Attribute Name	Data Type	Description
Location_id	Int (Primary Key)	Unique identifier assigned to each Location
Location_ name	Varchar (30)	Name of the Location

Table 3: Employees

Attribute Name	Data Type	Description
Employee_id	Int (Primary Key)	Unique identifier assigned to each Employees
Employee_ name	Varchar (30)	Name of the Location
Gender	Enum (M,F)	Employees's gender (Male or Female
Age	Int	Age of the employees in years
Hire-date	Date	Date of joining for Employees
Designation	Varchar (100)	Name of designation for Employees
Department_id	Int (Foreign Key)	Unique identifier assigned to each Department referenced to ForeignKey Constrint
Location_id	Int (Foreign Key)	Unique identifier assigned to each Locations referenced to ForeignKey Constrint
Salary	Decimal(10,2)	Employees Salary amount
Email	Varchar(100)	Email address for Employees

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4. Tools & Technologies

- **MySQL:** DDL Commands, My SQL Constraints

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5. Data Pre-Processing (MySQL)

Tasks Performed:

DDL Commands:

1. Database and Table Creation (CREATE):

Department table creation

```
create database if not exists employee_db;

use employee_db;

                                        -- Create three tables with appropriate data types and relationships
                                        -- 1. Department table creation

create table if not exists departments(
department_id int,
department_name varchar(100),

Primary Key (department_id)
);
desc departments;
                                        -- 2. Location table creation
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
department_id	int	NO	PRI	NULL	
department_name	varchar(100)	YES		NULL	

Location table creation

```
                                        -- 2. Location table creation

create table if not exists location(
location_id int primary key,
location_name varchar(30)
);
desc location;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
location_id	int	NO	PRI	NULL	
location_name	varchar(30)	YES		NULL	

Employees table creation

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```

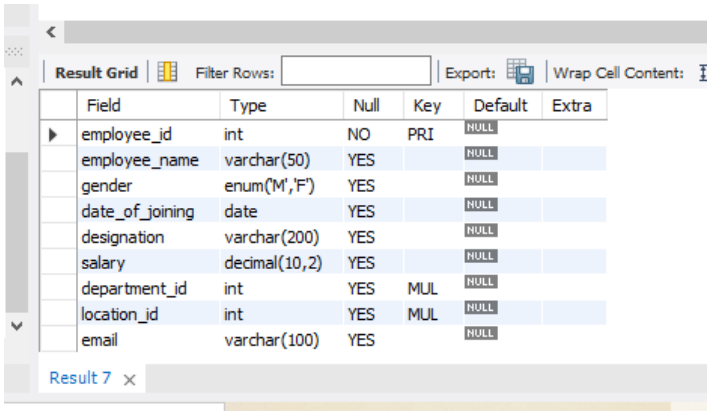
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	YES		NULL	
gender	enum('M','F')	YES		NULL	
age	int	YES		NULL	
hire_date	date	YES		NULL	
designation	varchar(100)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	

Result 6

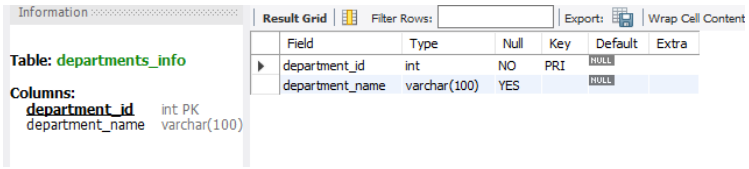
2. Table Alteration (ALTER)

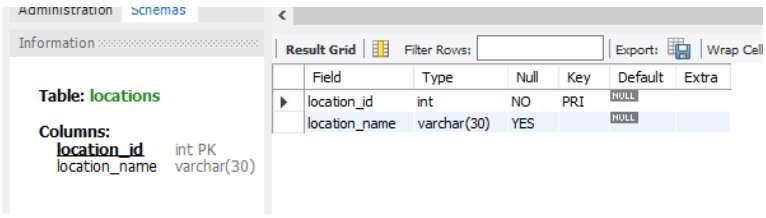
- Add a new column named "email" to the Employees table to store employee email addresses.
- Modify the data type of the "designation" column in the Employees table to support a wider range of values.
- Drop the “age” column from the Employees table.
- Rename the “hire_date” column to “date_of_joining”



3. Table Renaming (RENAME):

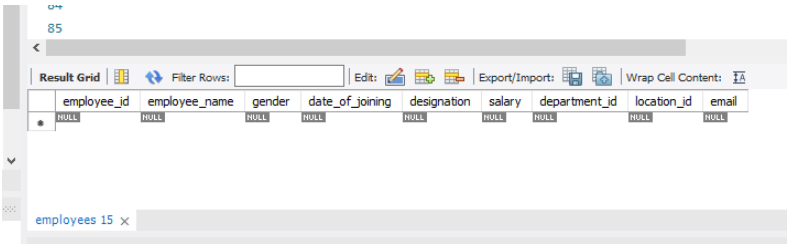
- Rename the "Departments" table to "Departments_Info"
- Rename the "Location" table to "Locations





4. Table Truncation (TRUNCATE):

- Truncate the Employees table



5. Database & Table Dropping (DROP):

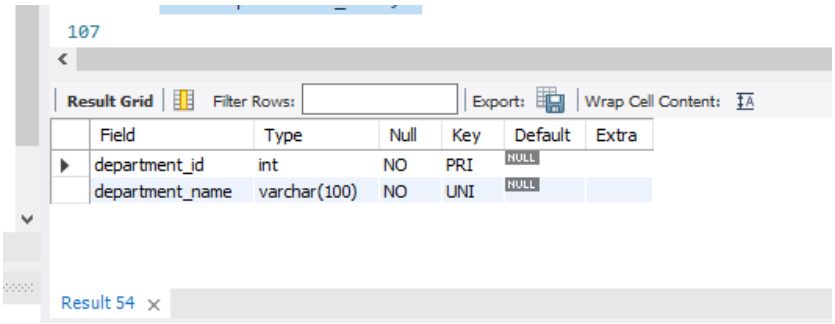
- Drop the Employees table and then the “employee” database



Constraints:

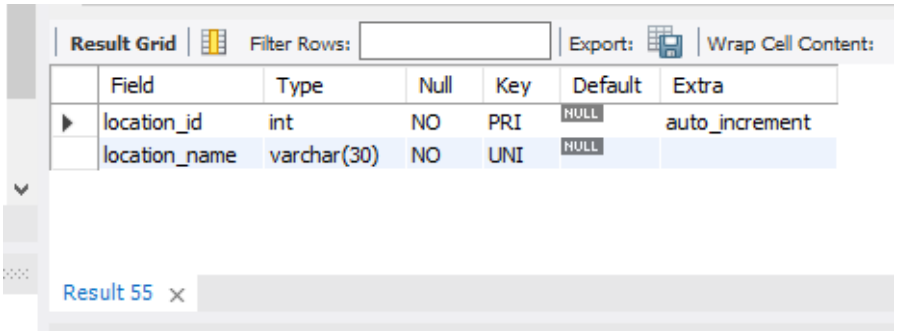
1.Departments Table:

- Ensure that the "department_id" uniquely identifies each department
- Set up constraints on the "department_name" to avoid duplicate and null entries



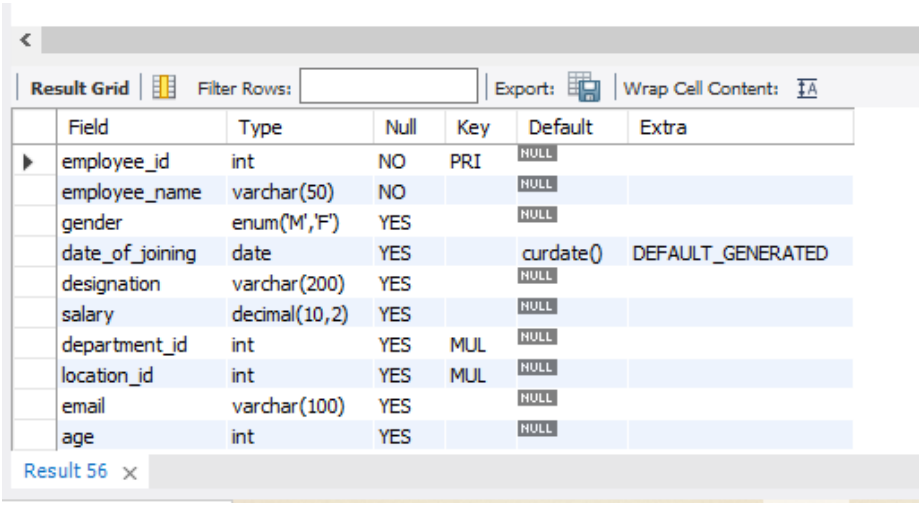
2.Locations Table:

- create incremented sequential for location_id
- Set up constraints on the "location_name" to avoid null and duplicate locations



3.Employees Table:

- Guarantee that each employee has a distinct identifier.
- Create a restriction to ensure that the employee's name is always provided.
- Limit the acceptable values for the "gender" field to only 'M' or 'F'.
- Enforce a condition to ensure that the employee's age is 18 or above.
- Automatically assign the current date to the "hire_date" field if not specified



Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
employee_name	varchar(50)	NO		NULL	
gender	enum('M','F')	YES		NULL	
date_of_joining	date	YES		curdate()	DEFAULT_GENERATED
designation	varchar(200)	YES		NULL	
salary	decimal(10,2)	YES		NULL	
department_id	int	YES	MUL	NULL	
location_id	int	YES	MUL	NULL	
email	varchar(100)	YES		NULL	
age	int	YES		NULL	

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6. Conclusions

This project provides a strong foundation in SQL database design, schema creation, and data validation, making it a valuable learning resource for understanding real-world database management systems and enterprise data organization.